

Technical Requirements — Satellite Imagery

TASCO Mapping Division · 2026-06 · Confidential · *Follows Part 1 — Introduction*

Area of Interest (AOI)

Our coverage requirements are calculated from a live H3 hexagonal grid model of Vietnam's road network and urban built-up areas. The interactive version is available at:

<https://lqtue.github.io/tasco-map-platform/sat-imagery/>

From that page you can:

- Visualise our target cells by type (urban / road corridor / combined)
- Adjust density and road-class thresholds to see how the envelope changes
- **Download the AOI** in two formats for your internal use:
 - **AOI Summary (JSON)** — total km² with tier and province breakdown, for pricing
 - **AOI Cells (GeoJSON)** — the exact H3 hex polygons, for archive inventory lookup

Priority tiers

Tier	Description	Approx. area
P1 — Urban built-up	Built-up zones around Vietnam's cities and towns	~15,355 km ²
P2 — Road corridors	10 km buffer along national roads (motorway, trunk, primary)	~5,662 km ²
P3 — Maritime territories	Strategic island territories (Trường Sa, Hoàng Sa archipelagos)	~867 km ²
Total initial envelope	Union of P1–P3 (overlaps removed)	~20,350 km ²

Geographic scope: Socialist Republic of Vietnam including recognised maritime territories.
CRS: WGS84 (EPSG:4326).

Note: Vietnam's monsoon season (May–October north; October–January south) produces heavy cloud cover simultaneously across much of the country. We expect multi-date mosaics; re-acquisition policy for cloud-rejected scenes is therefore a key evaluation criterion.

Image Specifications

Parameter	Requirement	Notes
Ground Sample Distance (GSD)	$\leq 0.5\text{ m}$	Sub-metre preferred
Spectral bands	RGB minimum; multispectral preferred	NIR useful for built-up separation
Cloud cover	$\leq 30\%$ per scene ($\leq 15\%$ preferred)	Cloud-free for island acquisitions
Off-nadir angle	$\leq 25^\circ$ preferred	Higher angles accepted for islands only
Geometric accuracy	CE90 $\leq 5\text{ m}$ absolute	Sufficient for OSM overlay
Acquisition age	≤ 18 months from delivery	Fresher preferred for P1 urban
Revisit / update	Minimum 1× per year per AOI	Quarterly preferred for urban zones

Delivery Requirements

Parameter	Requirement
Format	Cloud-Optimized GeoTIFF (COG)
Tiling	256 × 256 px or 512 × 512 px; overview levels included
Delivery mechanism	S3-compatible object storage
Catalog	STAC-compatible or manifest JSON per delivery batch
Metadata per scene	Acquisition date, cloud cover %, sensor ID, off-nadir angle
Lead time	Archive ≤ 5 business days; new tasking ≤ 8 weeks
Cloud QC	Scenes exceeding cloud cover threshold replaced at no additional charge

Licensing Requirements

We require:

1. Commercial use in our navigation products
 2. Right to create derivative works extracted from the imagery (vector data, map tiles, enriched road attributes)
 3. Redistribution of derivative products to end-users (consumers, fleet operators, B2B partners) without mandatory disclosure of the imagery source
 4. No mandatory attribution of the imagery provider in our end-user products
 5. Sub-licensing rights when supplying derived map data to third parties
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What We Ask From You

Please provide the following so we can complete our evaluation:

1. **Capability statement** — sensor fleet or aggregated sources, GSD range, revisit frequency over Vietnam, archive depth

2. **Archive inventory** — before committing to new tasking, please check the downloadable AOI GeoJSON against your archive at ≤ 0.5 m GSD and ≤ 18 months age and report coverage percentage by tier
 3. **Pricing proposal** — using the downloadable AOI Summary JSON as the basis:
 - Archive vs. new tasking price per km²
 - Re-acquisition cost if a scene is cloud-rejected
 - Minimum order and volume tiers
 - Long-term partnership or subscription pricing if available
 1. **Sample imagery** — one urban sample (~50 km², suggest central Hanoi or Ho Chi Minh City) and one road corridor sample (~100 km of QL1 central Vietnam) for pipeline evaluation before contract
 2. **Delivery timeline** — lead time for the initial envelope; guaranteed % of AOI delivered cloud-free; re-acquisition SLA
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Contact

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This brief is confidential and intended solely for the recipient organisation.
